

A data-curation study of U.S. state-level monitoring data, 2024–2025

Air Pollution Emissions & Honey Bee Colony Losses Across the United States

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🛡️ MOTIVATION

Why does this matter?

A drive down I-57 toward Chicago — endless cornfields stretching to the horizon — sparked a simple question:

How much of what we eat depends on something as fragile as a honey bee?

Recent USDA reports show record-high colony losses across 2024–2025.

Beyond the well-studied stressors — Varroa mites, pesticides, disease — environmental air quality has been flagged as a possible but under-quantified contributor.

75%

of food crops worldwide rely on honey bee pollination

Source: FAO

2024–25

marked record-high colony losses in the United States

Source: USDA NASS

5

major air pollutants tested across all U.S. states

Source: EPA AQS

What we set out to test

*Is there a statistically significant correlation between **air pollution emissions** in each U.S. state and **honey bee colony change rates** across 2024–2025?*

01

WHICH POLLUTANTS

Test five EPA-monitored pollutants — Ozone, PM2.5, CO, NO₂, SO₂ — measured at thousands of monitoring stations.

02

WHICH OUTCOMES

Three normalized colony metrics from USDA: Percent Lost, Percent Renovated, and Net Change Rate.

03

AT WHAT GRAIN

Pooled state × quarter panel (n ≈ 161–167) plus state-level averages (n = 42–44 states).

Two complementary federal datasets

DATASET 01		DATASET 02	
USDA NASS Honey Bee Colonies		EPA Air Quality System (AQS)	
COVERAGE	All 50 states · Jan 2024 – Apr 2025	COVERAGE	U.S. monitoring stations · 2024 – 2025
GRANULARITY	Quarterly · per-state colony counts	GRANULARITY	Daily site-level pollutant readings
FORMAT	Originally PDF · converted to CSV in Python	FORMAT	Pre-structured CSV files (downloaded direct)
KEY FIELDS	Colonies, Lost, Added, Renovated, % Lost	KEY FIELDS	Pollutant, Mean concentration, Obs. %, State
RECORDS	137 rows × 17 columns (post-cleaning)	RECORDS	~2.23 million rows × 10 columns
esmis.nal.usda.gov/publication/honey-bee-colonies		aq5.e5a.gov/aq5web/airdata/download_files.html	

Both datasets are public-domain, federally maintained, and merged at the State × Quarter level.

From raw release files to one merged panel

1

ACQUIRE

Download USDA PDF & EPA daily CSVs from official portals.

2

EXTRACT

Parse PDF tables into CSV with Python; load EPA CSVs with usecols filter.

3

CLEAN

Drop irrelevant stressors (Varroa, pesticides); coerce (X)/(Z) sentinels to NaN; standardize state names.

4

AGGREGATE

Roll EPA daily readings up to State × Quarter × Pollutant means; pivot to wide form.

5

MERGE

Inner-join bee panel with pollutant panel on (State, Quarter); drop "Other States" misalignment rows.

Output: a tidy panel of ~161–167 state-quarter observations linking colony health to pollutant exposure.

Why we used rates, not raw counts

California always dwarfs Connecticut in raw colony numbers. To compare states fairly, we normalized every bee outcome as a rate.

HONEY BEE OUTCOMES

Variable	Role
Percent Lost	Primary outcome — colony stress signal
Net Change Rate	(Added – Lost) / Max — population trajectory
Percent Renovated	Beekeeper response — re-queening intensity
Colonies, Max, Lost, Added	Dropped — size confound across states

AIR POLLUTANT EXPOSURE

- 
PM2.5
 Fine particulate matter — respiratory & olfactory disruption
- 
Ozone
 Ground-level O₃ — affects plant volatiles bees follow
- 
CO
 Carbon monoxide — combustion / urban gradient proxy
- 
NO₂
 Nitrogen dioxide — traffic & industry indicator
- 
SO₂
 Sulfur dioxide — coal/industrial point sources

Metric used: Arithmetic Mean (concentration) per pollutant.

◆ FINDING 01

PM_{2.5} — the most credible signal



THE HEADLINE

Higher PM_{2.5} → more colony loss.

$$r = +0.27$$

$p \approx 5 \times 10^{-4}$ · $n = 167$ state-quarters

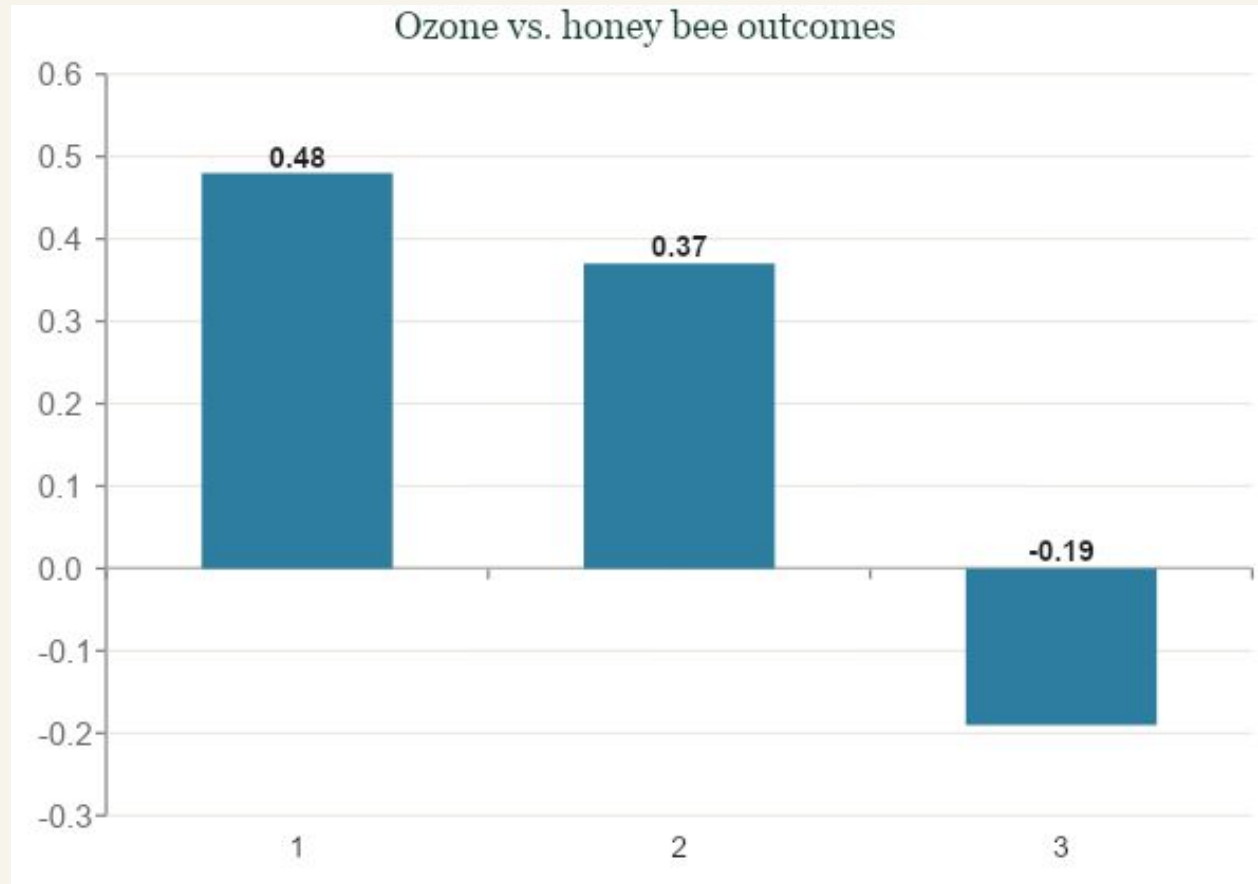
State-average: $r = +0.37$ ($p \approx 0.013$) · Spearman $\rho = +0.48$ ($p \approx 0.001$)

WHY WE TRUST IT

- Survives Bonferroni correction ($\alpha = 0.0033$).
- Consistent across pooled and state-average analyses.
- Biologically plausible — fine particles disrupt bees' olfactory navigation and respiratory function.

◆ FINDING 02

Ozone — strong, but probably confounded



THE PUZZLE

Higher ozone → larger net colony gain?

$$r = +0.48$$

$p \approx 4 \times 10^{-11}$ · Net Change Rate

LIKELY EXPLANATION

- Ozone peaks in warm, sunny southern/western states.
- Those same regions have longer foraging seasons & higher agricultural productivity.
- **The correlation likely captures climate, not a protective effect of ozone itself.**

FULL RESULTS

All 15 tests at a glance

	Ozone	PM2.5	CO	NO ₂	SO ₂
Percent Lost	-0.19	+0.27	+0.23	+0.18	+0.02
Percent Renovated	+0.37	+0.04	-0.05	+0.01	-0.03
Net Change Rate	+0.48	-0.20	-0.10	-0.06	+0.03

READING THE MATRIX

- Positive correlation (more pollutant → higher rate)
- Negative correlation (more pollutant → lower rate)
- Survives Bonferroni correction ($\alpha = 0.0033$)

3 of 15

tested correlations remain significant after Bonferroni correction. PM2.5 → Percent Lost is the most biologically credible.

Air quality matters — but it's only part of the picture.

WHAT WE FOUND

- PM2.5 is the only pollutant with a robust, biologically plausible link to colony loss.
- Ozone correlations are large but likely driven by climate and land-use confounds.
- CO and NO₂ show weak urbanization-shaped signals; SO₂ is essentially flat.

LIMITATIONS

- Only 6 quarters of data — limited statistical power for time-series tests.
- Equal-weighted state averages ignore apiary location and station coverage.
- Ecological correlations $\neq$ causation; many unmeasured confounders remain.

FUTURE WORK

- Extend to additional years as USDA & EPA release new data.
- Add covariates: temperature, drought, land use, neonicotinoid usage.
- Move from correlation to mixed-effects regression with state fixed effects.

Among 15 tested hypotheses, only PM2.5 → Percent Lost is consistent, significant, and biologically grounded. Cleaner air may mean healthier colonies.



Thank you!

